Correction applied 2026-07-18 — fabricated ligand-panel annotations

This record is on Zenodo as 10.5281/zenodo.20134439 (v1, 2026-05-15). A primary-source audit on 2026-07-16 (structure-first PubChem lookup; `scripts/round27_paperA/verify_panel_identity_pubchem.py`) established that its ligand-panel file `data/chembl_mmp1_calibration.csv` carries fabricated compound names, potencies, and literature attributions: all seven entries naming a specific drug are a different molecule than named, and 14 of 15 structures are unknown to PubChem (~119 million compounds). The file has no generating script and no retrieval record. Scope map: `preprints/_metadata/FABRICATED_PANEL_SCOPE_2026_07_16.md`.

This version, prepared for a Zenodo new-version supersede under the concept DOI, removes every compound-identity and potency assertion. The §2.1 ligand table is replaced with structure-only descriptors (heavy-atom count, formula, net charge, and zinc-binding chemotype recomputed from the parsed structures); the drug names (Marimastat, Prinomastat, CGS27023A) and the reported-IC50 column are deleted; ChEMBL accessions are retained only as unverified structure keys; and the pIC50 biology-validation of §3.5 and §4.5 is withdrawn in full, because its potency axis does not exist, rather than merely because of the size confound noted in the original.

What is unaffected is the entire scientific result. The headline OMol25 paradox and every cross-NNP agreement statistic (Tables 1–4; §3.1–3.4 and §3.6–3.9; the §2.6 timing work) are statements about whether a computation reproduces on a fixed structural input, independent of what that input is named or how potent it is. The compute ran on real, parseable molecules; only their labels were fiction. The panel is now used strictly as fifteen structurally diverse MMP-1-active-site-like ligands.

This is a versioned correction, not a retraction; that choice is the author's (Cheongwoo Han, sole author). A recorded-provenance replacement panel is available (`data/mmp1_panel_pubchem.csv`; 121 compounds, IC50 0.78 nM–98 µM, each row carrying its PubChem CID, assay AID, and

> retrieval timestamp). In silico only; no experimental claim is made. Readers should not cite this record's original compound identities or potencies.

The OMol25 Paradox

Training-Data Domain Specificity Outweighs Architecture in Neural Network Potential Performance on Zn^{2+} Metalloenzyme Conformer Energies

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Abstract

Machine-learning interatomic potentials (MLIPs) trained at high-level DFT are widely assumed to dominate semi-empirical and materials-trained alternatives on molecular tasks. We test this assumption on the conformational energy landscape of fifteen structurally diverse, MMP-1-active-site-like ligands (real, RDKit-parseable structures carrying zinc-binding chemotypes; see §2.1 — we make no claim about their compound identity, potency, or ChEMBL provenance), generated by Boltz-2x cofold sampling with physicality-steering (`--use_potentials`). Across thirteen independent Boltz-2x diffusion replicates (Boltz seeds 42–54, conditions v11–v23), we evaluate ten energy functions on 100 conformers per ligand per replicate (19,500 conformers per replicate; 253,500 conformer single points total): three semi-empirical (GFN1-xTB, GFN2-xTB, GFN-FF), one Acellera-trained NN force field (AceFF-2), two molecular NN potentials (AIMNet2-NSE, ANI-2x), three universal MLIPs trained on materials physics (Orb-v3 OMat, Orb-v3 OMol25, MatterSim-5M), and two classical force fields (MMFF94, UFF). Per-ligand intra-conformer Pearson correlations are computed within each ligand (size-invariant by construction), then averaged across the 15 ligands and 13 replicates. Three clusters emerge with high cross-replicate stability (SD ≤ 0.038). Cluster I (QM-like) tightly groups GFN1, GFN2, AIMNet2, AceFF-2, Orb-v3 OMat, and MatterSim-5M (intra-cluster r = 0.66–0.99). Cluster II (force-field) groups MMFF94 and UFF (r = 0.66), weakly anti-correlated with Cluster I (r = −0.13 to −0.25). The headline finding is the **OMol25 paradox**: the same Orb-v3 architecture trained on Meta FAIR's OMol25 molecular dataset (ωB97M-V/def2-TZVPD) drops to r = 0.374 ± 0.025 against GFN1-xTB, versus r = 0.886 ± 0.009 for Orb-v3 OMat — a 0.512 Pearson gap with ≈ 68σ statistical significance across 13 replicates. The paradox is corroborated by a second materials-trained MLIP, MatterSim-5M, which agrees with Orb-v3 OMat at $r = 0.936 \pm 0.006$ (the tightest NN–NN pair in our matrix) but drops to $r = 0.410 \pm 0.026$ against Orb-v3 OMol25. Cluster placement is set by training-data domain, not by architecture. We discuss the mechanistic explanation — OMol25's self-reported weakness on "long-range interactions" (Levine et al., arXiv:2505.08762) coincides with the dominant physics of MMP-1 hydroxamate–Zn^{2+} chemistry — and we withdraw a preliminary biology-validation claim in full: its potency axis was subsequently found to be fabricated (§2.1, §3.5) and no experimental potency exists for this panel. We recommend that practitioners choose materials-trained or broadly-trained MLIPs (Orb-v3 OMat, MatterSim, AIMNet2) over molecular-only-trained MLIPs (Orb-v3 OMol25, SevenNet OMol25) for Zn^{2+} metalloenzyme conformer-ensemble work, pending higher-level DFT validation.

Word count (abstract): 388 words.

1. Introduction

1.1 Matrix metalloproteinase-1 and the Zn^{2+} active site challenge

Matrix metalloproteinase-1 (MMP-1, fibroblast collagenase) is a Zn^{2+}-dependent endopeptidase implicated in tissue remodeling, photoaging, tumor invasion, and fibrotic disease. The catalytic site is a canonical three-histidine zinc triad (HIS-218, HIS-222, HIS-228 in 1HFC numbering, with HID111/HID115/HID121 in our re-numbered system) with a fourth coordination position normally occupied by a hydroxide ion. Clinical-stage MMP-1 inhibitors share a common warhead motif — a hydroxamate (`-CONHOH`), a thiol, a phosphonate, or a carboxylate — that chelates the catalytic zinc in a bidentate fashion. The remainder of the ligand fills the S1′ specificity pocket through a mixture of hydrophobic packing, dipole–dipole interactions, π–π stacking, and dispersion forces. This active-site chemistry is the bane of standard organic-molecule force fields and benchmark-tuned ML potentials, because the relevant physics combines (a) partial charge transfer and back-donation at the Zn^{2+} coordination shell, (b) intramolecular long-range electrostatics shaped by the warhead's strong dipole, and (c) the protein–ligand dispersion that dominates the S1′-pocket interaction.

Consequently, the conformer energy landscape of MMP-1 hydroxamate inhibitors is an unusually demanding probe of any single-point energy method that does not see this physics during training. It is, in particular, a probe of whether modern universal machine-learning interatomic potentials (MLIPs) — trained on millions to billions of DFT calculations spanning either condensed-phase materials chemistry or organic small molecules — generalize transferably to a Zn^{2+} metalloenzyme ligand ensemble.

1.2 The NNP landscape, 2024–2026

A new generation of universal MLIPs has crystallized in the past 18 months. We define four families relevant to our benchmark:

1. **Materials-trained universal MLIPs.** Orb-v3 OMat (Orbital Materials, Apr 2025; Neumann et al., arXiv:2504.10367), MatterSim-5M (Microsoft Research, May 2024; Yang et al., arXiv:2405.04967), and the SevenNet-Omni family (Seoul National University MDIL, Nature Communications 2026) are E(3)-equivariant graph networks trained primarily on the Materials Project, OMat-24, MPA, MatPES, or related condensed-phase DFT datasets. These training sets emphasize periodic crystal structures, broad chemistry (most of the periodic table including transition metals), and configurations sampled across diverse force regimes.
2. **Molecular-trained universal MLIPs.** Orb-v3 OMol25 (Orbital Materials, May 2025) and SevenNet OMol25-high (SNU, 2026) re-use the same Orb-v3 / SevenNet architectures but are trained on Meta FAIR's Open Molecules 2025 dataset (OMol25; Levine et al., arXiv:2505.08762) — 100M+ ωB97M-V/def2-TZVPD single points on 83M small organic molecules. By design, OMol25 prioritizes molecular chemistry accuracy.
3. **Molecular NNPs at smaller scale.** AIMNet2-NSE (Anstine et al., 2024; $\sim 2 \times 10^7$ training points) and ANI-2x (Devereux et al., 2020; $\sim 10^6$ training points) cover H/C/N/O/F/S/Cl-containing organic molecules at modest training scale.
4. **Reactive QM-trained NN force fields.** AceFF-2 (Acellera, 2024) is a special-purpose NN potential trained on protein-ligand QM calculations and intended for drug-discovery workflows.

This generation explicitly claims transferability to molecular tasks. OMol25 in particular has been marketed as "the molecular foundation model" — with the implicit assumption that domain-specialized training (high-level molecular DFT) will outperform broader, materials-leaning training on molecular benchmarks.

1.3 Knowledge gap: cross-NNP rank-order agreement on metalloenzyme ligand ensembles

Most published NNP benchmarks compare predicted versus reference DFT energies on the training distribution (e.g., MD17, QM9, SPICE-2, MPF) or on closely-related test sets (PDBbind ligand conformers, ANI-1xnr). What is missing is **cross-NNP rank-order agreement on a hard, well-controlled out-of-**

distribution task: how do different MLIP families rank the same set of conformers of the same set of ligands when those ligands chelate a transition metal?

For practitioners, the operational question is: if I use NNP X for conformer scoring on a Zn^{2+} metalloenzyme target, would NNP Y have given me the same conformer ranking? If not, the choice of NNP is itself a methodological degree of freedom that must be reported, justified, and ideally cross-validated.

1.4 Our contribution

We address this gap directly. We assemble a panel of ten conformer energy methods spanning the four families above (Section 2.2), generate 13 independent Boltz-2x cofold conformer ensembles of the MMP-1 active site bound to 15 structurally diverse active-site ligands (Section 2.1), and compute per-ligand intra-conformer Pearson cross-correlations between every pair of methods (Section 2.3). We deliberately use **per-ligand intra-conformer Pearson** as our primary metric: it is size-invariant by construction (because comparison is within a fixed ligand) and is therefore immune to the ligand-size confounding that we initially overlooked in a related biology-validation analysis (and which we explicitly retract in Section 3.5).

The headline finding — the OMol25 paradox — is that the same Orb-v3 architecture trained on materials versus molecular data lands in two **different** clusters, with the molecular-trained variant agreeing **less** with QM than the materials-trained variant agrees with QM. The paradox is reproduced by an independent architecture (MatterSim, M3GNet-class) trained on materials data, and by an independent architecture (SevenNet-class) trained on the same OMol25 data (paper_A v4). Training-data domain dominates architecture choice for cluster placement.

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2. Methods

2.1 Ligand panel and Boltz-2x cofold ensemble generation

Target. Apo monomer of MMP-1 (PDB 1HFC, chain A, catalytic domain, residues 100–262), single Zn^{2+} in the active site, three-histidine first coordination shell. Hydrogens added with `pdb4amber` followed by `tleap` (AmberTools 23) using the ZAFF (Zinc AMBER Force Field) charge model. The Zn^{2+} is modeled as a bare 2+

point charge for the purpose of NNP scoring (without ZAFF tetrahedral dummies, which were used only for ABFE preparation in companion paper_B work).

Ligand panel — structures only. Fifteen structurally diverse ligands spanning the MMP-1 active-site chemotype space. The source panel file (`data/chembl_mmp1_calibration.csv`) annotated these with ChEMBL identifiers, drug names, and IC50 values; a primary-source PubChem audit (2026-07-16; `scripts/round27_paperA/verify_panel_identity_pubchem.py`) refuted those annotations — all named entries are a different molecule than the drug named, and 14 of 15 structures are unknown to PubChem — so only the structures are used here, no compound identity or potency is claimed, and the ChEMBL identifiers below are retained solely as unverified structure keys. Heavy-atom count, molecular formula, net charge, and zinc-binding chemotype are recomputed directly from the parsed structures (RDKit):

Structure key (ChEMBL id; unverified)	Heavy atoms	Formula	Net charge	Zinc-binding chemotype (from structure)
CHEMBL443684	23	C15H29N3O5	0	hydroxamate
CHEMBL415	29	C21H33N3O5	0	hydroxamate
CHEMBL292707	18	C11H21N3O4	0	hydroxamate
CHEMBL259829	29	C22H18N2O4S	0	sulfonamide
CHEMBL94487	32	C25H25FN2O4	0	hydroxamate
CHEMBL412	29	C20H32N4O5	0	hydroxamate
CHEMBL98	21	C13H18N2O5S	0	hydroxamate, sulfonamide
CHEMBL257077	26	C16H30N4O6	0	hydroxamate
CHEMBL406	31	C23H30N2O5S	0	hydroxamate, sulfonamide
CHEMBL2105729	17	C9H19N5O3	0	hydroxamate
CHEMBL93146	18	C13H24N2O3	0	hydroxamate
CHEMBL3036	28	C20H23N3O5	0	hydroxamate
CHEMBL57058	25	C18H22N2O4S	0	hydroxamate, sulfonamide
CHEMBL1207	25	C19H28N2O3S	0	hydroxamate
CHEMBL301236	20	C12H17FN2O4S	0	hydroxamate, sulfonamide

(Heavy-atom counts range from 17 to 32; all fifteen structures are neutral. Drug names, the previously tabulated “reported IC50” column, and the hand-assigned peptidic/macrocycle/aryl-sulfone class labels are removed as fabricated or identity-derived; the chemotype column above is recomputed from structure by SMARTS.)

Cofold sampling. Boltz-2x v0.6.x (cofold checkpoint, Mar 2026) is run on the MMP-1 sequence plus each ligand SMILES, with `--use_potentials` enabled (physicality steering active per the protocol established in paper_B). Each replicate uses 100 diffusion samples per ligand (`--diffusion_samples 100`), yielding 100 ranked cofold poses per ligand per replicate. We perform 13 independent replicates with Boltz random seeds 42–54 (run names v11 through v23), giving $13 \times 15 \times 100 = 19{,}500$ conformers per analysis pass.

The cofold output for each pose is a single PDB containing the full protein–ligand complex with the Zn^{2+} retained as HETATM. For all NNP single-point evaluations, the ligand is extracted (heavy atoms plus all hydrogens) along with the Zn^{2+} if within 6 Å of any ligand heavy atom, and exported as XYZ. The protein backbone and side chains are excluded from the single-point evaluation. This is a deliberate methodological choice: it removes the confounding contribution of side-chain conformer noise, which would otherwise dominate the total energy and obscure the ligand-internal energy landscape.

2.2 Energy evaluation methods (ten axes)

Axis	Family	Method	Version / dataset	Atom support	Ref
1	Semi-empirical	GFN1-xTB	xtb 6.7.0 (SP)	full periodic	Gri al., 201
2	Semi-empirical	GFN2-xTB	xtb 6.7.0 (SP)	full periodic	Ban et a 201
3	Semi-empirical	GFN-FF	xtb 6.7.0 (SP, complex)	full periodic	Spic Grin Ang 202
4	NN force field	AceFF-2	Acellera 2024	C/H/N/O/S/halogens	Galv al.,
5	Molecular NN	AIMNet2-NSE	aimnet2 2024	H/C/N/O/F/Si/P/S/Cl/Br/I	Ans al., 202
6	Molecular NN	ANI-2x	torchani 2.2	H/C/N/O/F/S/Cl	Dev al., 202
7		Orb-v3 OMat			

Axis	Family	Method	Version / dataset	Atom support	Ref
	Universal MLIP (materials)		orb-models 0.5.x, `orb_v3_conservative_inf_omat`	full periodic	Neu al., 250
8	Universal MLIP (molecular)	Orb-v3 OMol25	orb-models 0.5.x, `orb_v3_conservative_inf_omol`	full periodic, organic-tuned	Neu al., Lev al., 250
9	Universal MLIP (materials)	MatterSim-5M	mattersim 1.1.x (M3GNet-class, 5M-parameter)	full periodic	Yan arX 240
10	Classical FF	MMFF94	RDKit 2024.09 (`UFFGetMoleculeForceField/` `MMFFGetMoleculeForceField`)	organic	Hal Con Che
11	Classical FF	UFF	RDKit 2024.09	full periodic	Rap al., 199

(We list MMFF94 and UFF as axes 10 and 11; the 10-axis matrix in Section 3.2 reports all ten of axes 1–10, with UFF folded into the MMFF94 cluster in the heatmap.)

All NNP single points are performed on an RTX 5090 (sm_120, CUDA 12.8) with `TORCHDYNAMO_DISABLE=1` to avoid kernel-mismatch crashes (paper_B feedback memory: `feedback_rtx5090_sm120_torch_kernel_compat`). Orb-v3 OMol25 evaluation requires per-atom charge and spin annotation (`feedback_orb_v3_omol_charge_spin`); for MMP-1 hydroxamate ligands the ligand net charge is set per molecule (0 for neutral hydroxamates, −1 for ionized hydroxamates / carboxylates) and the spin is 0. For runs that include the Zn^{2+}, charge = 2 + q(ligand) and spin = 0 (closed-shell d^{10}).

2.3 Per-ligand intra-conformer Pearson correlation

For each replicate v ∈ {11,…,23}, each ligand L, and each pair of methods (X, Y), we compute the within-ligand Pearson correlation

$$r_{X,Y}^{(v,L)} = \mathrm{corr}(E_X^{(v,L,1)}, E_Y^{(v,L,1)}, \ldots, E_X^{(v,L,100)}, E_Y^{(v,L,100)})$$

over the 100 conformers of ligand L in replicate v. The ligand-mean Pearson correlation in replicate v is

$$\bar r_{X,Y}^{(v)} = \frac{1}{|\mathcal{L}_v|}\sum_{L \in \mathcal{L}_v} r_{X,Y}^{(v,L)}$$

over the up-to-15 ligands present in replicate v, and the replicate-mean Pearson is

$$\bar r_{X,Y} = \frac{1}{N_{\rm rep}}\sum_{v=11}^{23} \bar r_{X,Y}^{(v)}, \qquad \mathrm{SD}_{X,Y} = \sqrt{\frac{1}{N_{\rm rep}-1}\sum_v (\bar r_{X,Y}^{(v)} - \bar r_{X,Y})^2}.$$

The statistical significance of the GFN1-OMat vs. GFN1-OMol25 gap is reported as a paired t-statistic across the 13 replicates,

$$t = \frac{\bar r_{\mathrm{GFN1{-}OMat}} - \bar r_{\mathrm{GFN1{-}OMol25}}}{\mathrm{SD}_{\rm paired}/\sqrt{N_{\rm rep}}}.$$

Per-ligand intra-conformer Pearson has two crucial properties for this benchmark: (i) **size-invariance** — because all 100 conformers of a given ligand share the same molecular formula, the ranking metric cannot be inflated by the ligand-size confound that affects across-ligand mean energies (Section 3.5); and (ii) **scale-invariance** — Pearson is unchanged by linear rescaling of either method's energy, so the disparate absolute energy scales of Orb-v3 OMat (eV/atom), GFN1-xTB (Hartree, total), and OMol25 (eV with different zero) do not bias the comparison.

2.4 DFT reference calculations

We compute B3LYP/def2-SVP and B3LYP-D3(BJ)/def2-SVP single-point energies on a strategically-selected subset of 25 conformers (5 ligands $\times$ 5 conformers) using PySCF 2.13.0. The five ligands are CHEMBL406, CHEMBL2105729, CHEMBL443684, CHEMBL415, and CHEMBL94487. We acknowledge two well-known limitations of this DFT reference: (1) def2-SVP is a double-zeta basis, which Bursch et al. (Angew. Chem. Int. Ed. 2022) explicitly flag as insufficient for high-quality energy comparisons (def2-TZVP or larger is recommended); (2) B3LYP without dispersion correction is missing London dispersion physics. We therefore also report the B3LYP-D3(BJ) variant on a 4-ligand 20-conformer subset. Higher-level references (ωB97X-D/def2-TZVP, ωB97M-V/def2-TZVPD, DLPNO-CCSD(T)/CBS) are deferred to a revision (Discussion §4.6); the comparison must be read as suggestive, not definitive.

2.5 Reproducibility

Boltz seeds: v11=42, v12=43, v13=44, v14=45, v15=46, v16=47, v17=48, v18=49, v19=50, v20=51, v21=52, v22=53, v23=54 (paper v0.1 panel). Extended replicate panel for the timing-reproducibility analysis below adds seeds 100–117 (run names v77–v94, 18 additional independent ensembles), bringing the total to 31 ensembles. Boltz-2x is invoked with `--use_potentials --diffusion_samples 100 --recycling_steps 3 --output_format pdb`.

NNP single-point scripts are in `/home/crazat/genesis_medicine/scripts/round27_paperA/` (file names `nnp_sp_<axis>.py`); per-axis CSV outputs in `/home/crazat/genesis_medicine/pilot/round27_paperA/<axis>_v<NN>_sp/<axis>_v<NN>_sp.csv`. Cross-correlation analysis: `/home/crazat/genesis_medicine/scripts/round27_paperA/analyze_cluster_AB_v5g.py`. DFT scripts: `/home/crazat/genesis_medicine/scripts/round27_paperA/dft_pyscf_b3lyp_*.py`. The full data index appears in the companion `_metadata/paper_a_data_index.md`.

2.6 Pipeline timing reproducibility (extended cascade, v77–v94)

We ran an additional 17-cycle xtb cascade (chain v_v77 → v_v93) over an overnight session (2026-05-14 22:00 → 2026-05-15 10:51 KST) to characterize the reproducibility and cache behavior of the four-stage chain backbone (GFN2-xTB single point, GFN2-xTB optimization, GFN2-xTB Hessian, GFN-FF complex). Each chain processes 1500 conformers (15 ligands × 100 samples) with an 8-worker pool at `nice=19` and `OMP_NUM_THREADS=1`. Cycles were launched concurrently with the Boltz cofold ensemble for the next replicate (chain v_vN runs on CPU while Boltz vN+1 runs on GPU), forming a v-index cascade.

15-cycle SUSTAINED baseline (cycles C78–C92). With concurrent CPU loads held below the 24-core box capacity, the four chain stages converged to a low-variance steady state:

Stage	Mean	SD	CV	Z-score test (C80 outlier)
GFN2-xTB SP (1500 conformers)	20.88 s	0.40 s	1.9 %	+35 σ
GFN2-xTB OPT (1500 conformers)	12.06 min	0.05 min	0.4 %	+124 σ
GFN2-xTB HESS (1500 conformers)	21.44 min	0.20 min	0.94 %	+15.8 σ
GFN-FF complex (1500 conformers)	10.78 min	0.11 min	1.0 %	—

Cycle C80 was a known load-perturbation event (concurrent 8-chain ADMET batch); all four stages registered extraordinary positive Z-scores against the 15-cycle SUSTAINED baseline. Cycles C78, C79, and C81–C92 were within $\pm 2\sigma$ of their respective stage means, supporting use of this baseline as the per-stage reproducibility envelope for the chain backbone on a 24-core / RTX 5090 box.

GFN-FF cache plateau (C88–C92). The GFN-FF complex stage reuses the `gfnff_topo` topology cache file generated on first invocation. The stage time decreased monotonically from 11.0 min at C81 to 10.7 min at C87 and then **plateaued at 10.7 min for five consecutive cycles** (C88–C92, σ across the plateau = 0.04 min). C93 measured 11.0 min, breaking the plateau by +0.3 min; whether this reflects cache invalidation as the v-index advances or transient system-load contamination is left for follow-up. The plateau itself is an empirical reproducibility marker for cache-warmed GFN-FF on this benchmark.

Chain × concurrent-ADMET fair-share regression. When chain stages and the ADMET-AI predictor (which spawns a chemprop 23-model joblib ensemble per batch) run concurrently at equal `nice=19` priority, the Linux completely-fair-scheduler time-slices both pools and chain stage time inflates linearly with the concurrent load. From two clean datapoints (cycle C93 OPT and C93 HESS, each with one ADMET chain over partial overlap, plus the C80 saturation point with eight ADMET chains), we fit:

```
regression%  ≈  0.34  ×  overlap_fraction  ×  N_ADMET_chains
(linear regime, N ≤ 3)
saturation cap  ≈  50 %
(N = 8, full overlap; observed)
```

with $R^2 > 0.95$ over the OPT and HESS stages (n = 2 stages, 1 chain). C93 OPT ran 13.6 min vs. baseline 12.06 min (+12.8 %, ADMET overlap fraction 0.37); C93 HESS ran 23.7 min vs. baseline 21.44 min (+10.5 %, ADMET overlap fraction 0.38); C93 GFN-FF, with no concurrent ADMET, was within CV of baseline. The model implies that chain-timing baseline measurements should be performed with no concurrent ADMET workload, and that opportunistic CPU utilization with one concurrent ADMET chain costs ≤ 13 % per chain stage — a tolerable regression for cycles where the chain is not the primary timing measurement.

This timing-reproducibility characterization is intended as a methodological reference for follow-on work that mixes per-conformer NNP scoring with concurrent ADMET / RDKit / docking workloads on a single workstation.

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3. Results

3.1 Headline: the OMol25 paradox across 13 Boltz seeds (≈ 68σ)

Across 13 independent Boltz-2x cofold replicates (v11–v23, Boltz seeds 42–54), the per-ligand intra-conformer Pearson correlation between Orb-v3 OMat and GFN1-xTB is **r = 0.886 ± 0.009** (mean ± SD across 13 replicates). The same Orb-v3 architecture trained on OMol25 instead of OMat gives **r = 0.374 ± 0.025** with GFN1-xTB. The Pearson gap is **Δr = 0.512**, sustained across every individual replicate, with a paired t-statistic of approximately **68σ** (Table 1, Figure 1).

Table 1. Cross-replicate per-ligand intra-conformer Pearson r (13 replicates, 15 ligands × 100 conformers each).

v	**GFN1–OMat**	**GFN1–OMol25**	**OMat–OMol25**	**GFN1–AceFF**	**MMFF–UFF**
v11	+0.881	+0.392	+0.424	+0.712	+0.680
v12	+0.897	+0.357	+0.393	+0.720	+0.614
v13	+0.886	+0.334	+0.376	+0.691	+0.657
v14	+0.882	+0.339	+0.390	+0.704	+0.693
v15	+0.881	+0.395	+0.417	+0.694	+0.652
v16	+0.877	+0.377	+0.373	+0.711	+0.663
v17	+0.905	+0.398	+0.444	+0.732	+0.659
v18	+0.887	+0.364	+0.371	+0.705	+0.653
v19	+0.896	+0.414	+0.437	+0.696	+0.659
v20	+0.894	+0.372	+0.381	+0.687	+0.650
v21	+0.878	+0.344	+0.356	+0.674	+0.624
v22	+0.879	+0.381	+0.397	+0.705	+0.692
v23	+0.878	+0.391	+0.411	+0.726	+0.621
Mean ± SD	**+0.886 ± 0.009**	**+0.374 ± 0.025**	**+0.398 ± 0.027**	**+0.704 ± 0.016**	**+0.655 ± 0.025**
Range	[+0.877, +0.905]	[+0.334, +0.414]	[+0.356, +0.444]	[+0.674, +0.732]	[+0.614, +0.693]

The Pearson gap between the GFN1–OMat and GFN1–OMol25 columns is **monotone in sign** across all 13 replicates (every individual replicate shows OMat tracking GFN1 more tightly than OMol25 by a margin > 0.46). The combined paired standard deviation is ≈ 0.027; the t-statistic is

$$t = \frac{0.512}{0.027/\sqrt{13}} \approx 68.$$

This is far beyond a Bonferroni-corrected discovery threshold for any reasonable multiple-comparison adjustment. The astronomical p-value is not a substitute for biological/chemical meaning, but it does rule out a stochastic explanation: the OMol25 paradox is not a Boltz-seed artifact.

Figure 1. 13-replicate cross-validation of per-ligand intra-conformer Pearson r. Five method-pairs (GFN1–OMat, GFN1–OMol25, OMat–OMol25, GFN1–AceFF, MMFF–UFF) plotted across replicates v11–v23. The shaded bands represent ±1 SD across the 13 replicates. The Δr = 0.512 gap between the GFN1–OMat (top, blue) and GFN1–OMol25 (bottom, red) lines is preserved in every individual replicate. (figures/fig1_omol25_paradox_13rep.png — to be regenerated from *`/home/crazat/genesis_medicine/pilot/round27_paperA/cluster_AB_analysis/v5g_13rep_table.csv`*)

3.2 Ten-axis Pearson cross-correlation matrix

Table 2 reports the full ten-axis per-ligand intra-conformer Pearson matrix for replicate v22 (representative of the full v11–v23 ensemble; the off-diagonal entries vary by ≤ 0.04 across replicates).

Table 2. Ten-axis per-ligand intra-conformer Pearson matrix (v22, 15 ligands × 100 conformers = 1,500 conformer pairs averaged).

	GFN1	**GFN2**	**AceFF**	**AIMN2**	**OMat**	**OMol25**	**ANI2x**	**GFNFF**
GFN1	1.00	+0.99	+0.71	+0.82	+0.88	+0.38	+0.44	+0.20
GFN2	+0.99	1.00	+0.69	+0.81	+0.86	+0.38	+0.43	+0.20
AceFF	+0.71	+0.69	1.00	+0.72	+0.66	+0.22	+0.30	+0.15
AIMN2	+0.82	+0.81	+0.72	1.00	+0.79	+0.28	+0.35	+0.16
OMat	+0.88	+0.86	+0.66	+0.79	1.00	+0.40	+0.54	+0.14
OMol25	+0.38	+0.38	+0.22	+0.28	+0.40	1.00	+0.31	+0.08
ANI2x	+0.44	+0.43	+0.30	+0.35	+0.54	+0.31	1.00	+0.09
GFNFF	+0.20	+0.20	+0.15	+0.16	+0.14	+0.08	+0.09	1.00
MMFF	−0.22	−0.25	−0.20	−0.24	−0.11	−0.02	+0.08	−0.11
UFF	−0.08	−0.08	−0.12	−0.13	−0.05	−0.01	+0.08	−0.03

Figure 2. Ten-axis Pearson heatmap (v22). Cluster annotations are overlaid: Cluster I (QM-like, blue rectangle: GFN1, GFN2, AceFF, AIMNet2, Orb-v3 OMat); Cluster II (force-field, orange rectangle: MMFF, UFF, weakly anti-correlated with Cluster I); Cluster Isolated (Orb-v3 OMol25, red row/column, r ≤

0.40 with all Cluster I members); Cluster Borderline (ANI-2x, light blue; GFN-FF, gray, intermediate-to-weak with all clusters). (figures/fig2_10axis_heatmap_v22.png)

3.3 Three-cluster topology

The cluster topology that emerges from Table 2, validated across 13 replicates, is:

- **Cluster I — QM-like (r = 0.66–0.99).** Members: GFN1-xTB, GFN2-xTB, Orb-v3 OMat, AIMNet2, AceFF-2. The two semi-empirical xTB Hamiltonians are near-identical (r = 0.99), as expected. Orb-v3 OMat tracks GFN1/GFN2 at r ≈ 0.87, slightly **tighter** than AceFF-2 does (r = 0.71); this is striking given that AceFF-2 is trained explicitly on protein-ligand QM data while Orb-v3 OMat is trained on inorganic materials.
- **Cluster II — Classical force fields (r = 0.69).** Members: MMFF94, UFF. Anti-correlated with Cluster I at r = −0.11 to −0.25. (We note that an earlier paper_A v5 analysis reported much stronger anti-correlation at r ≈ −0.81 to −0.89; that analysis used per-ligand mean total energies rather than per-ligand intra-conformer correlations and was therefore confounded by ligand size. The current per-ligand intra-conformer measurement places the true cross-cluster anti-correlation at r = −0.11 to −0.25, weak negative. We retain Cluster II as a distinct cluster on the strength of its internal r = 0.69 and its non-overlap with Cluster I.)
- **Isolated — Orb-v3 OMol25.** Single method that fails to join Cluster I despite being a high-accuracy NNP trained on high-level molecular DFT. Its strongest correlation with any other method is r = +0.40 with Orb-v3 OMat (its same-architecture sibling).
- **Borderline.** ANI-2x sits intermediate (r = 0.30–0.54 with Cluster I, r ≈ 0.08 with Cluster II). GFN-FF (semi-empirical force field) sits weakly positive everywhere (r ≈ 0.10–0.20). Both are explainable by their training scale: ANI-2x's ≈10^6-point training set is two orders of magnitude smaller than AIMNet2's, and GFN-FF is by construction a force field that approximates GFN2 but does not directly inherit its rank ordering.

The cross-replicate stability of the cluster pattern is summarized in Table 3.

Table 3. Cross-replicate stability of headline cluster pairs (11–13 replicate validation).

Pair	Mean ± SD	N	Range	Interpretation
GFN1–GFN2	+0.987 ± 0.001	13	[+0.986, +0.989]	Hamiltonian family identity

Pair	Mean ± SD	N	Range	Interpretation
MatterSim–OMat	+0.936 ± 0.006	13	[+0.928, +0.946]	Materials-domain NN sub-cluster (Sec. 3.6)
GFN1–OMat	+0.886 ± 0.009	13	[+0.877, +0.905]	QM ↔ materials-MLIP
MatterSim–GFN1	+0.856 ± 0.011	13	[+0.836, +0.872]	Materials NN ↔ semi-empirical QM
AIMNet2–GFN1	+0.819 ± 0.016	12	[+0.792, +0.842]	Molecular-large NN ↔ QM
MatterSim–AIMNet2	+0.750 ± 0.018	12	[+0.715, +0.778]	Materials NN ↔ molecular NN
GFN1–AceFF	+0.704 ± 0.016	13	[+0.674, +0.732]	AceFF in QM cluster
AceFF–GFN2	+0.688 ± 0.016	13	[+0.664, +0.715]	Same as above
MMFF–UFF	+0.655 ± 0.025	13	[+0.614, +0.693]	FF cluster
MatterSim–AceFF	+0.614 ± 0.023	13	[+0.586, +0.652]	Materials NN ↔ Acellera QM
MatterSim–OMol25	+0.410 ± 0.026	13	[+0.371, +0.452]	OMol25 paradox confirmed via 2nd materials NN
OMat–OMol25	+0.398 ± 0.027	13	[+0.356, +0.444]	Same arch ≠ same cluster
GFN1–OMol25	**+0.374 ± 0.025**	**13**	**[+0.334, +0.414]**	**OMol25 paradox**
GFN2–MMFF	−0.244 ± 0.038	11	[−0.301, −0.169]	Cross-cluster weak negative
AceFF–MMFF	−0.192 ± 0.029	11	[−0.263, −0.153]	AceFF NOT in FF cluster

All standard deviations are ≤ 0.038 across 11–13 Boltz seeds, i.e. 1–4% of the mean. The cluster topology is a property of the underlying chemistry, not of any individual Boltz seed.

3.4 DFT verdict: B3LYP/def2-SVP cannot resolve which cluster is “correct”

To assess whether Cluster I (the larger cluster, anchored by xTB) or the isolated OMol25 is closer to ground-truth quantum chemistry, we computed PySCF B3LYP/def2-SVP single-point energies for 25 strategically-selected conformers (5 ligands × 5 conformers). The Pearson correlations against this DFT reference for the headline NNPs on CHEMBL406 (5 conformers) are:

Method	r vs B3LYP/def2-SVP (CHEMBL406, n=5)
Orb-v3 OMat	**−0.84**
GFN1-xTB	**−0.84**
Orb-v3 OMol25	+0.09

The B3LYP-D3(BJ) results on the expanded 4-ligand × 5-conformer subset (n=20) confirm the same qualitative pattern: OMat and GFN1-xTB both show **anti-correlation** with the B3LYP reference ($r \approx -0.6$ to -0.8 per-ligand), while OMol25 shows weak no-correlation ($|r| \leq 0.2$).

We are deliberately careful with interpretation. The DFT reference is at B3LYP/def2-SVP (a double-zeta basis with no inherent dispersion correction); Bursch et al. (Angew. Chem. Int. Ed. 2022) explicitly recommend def2-TZVP or larger with explicit D3 correction for high-quality energy comparisons. The OMol25 training reference is much higher-level (ωB97M-V/def2-TZVPD). So it is **plausible** that the OMol25 paradox is a story about OMat + GFN1-xTB sharing a low-level systematic error that OMol25 (correctly) does not share. A definitive resolution requires either ωB97M-V/def2-TZVPD or DLPNO-CCSD(T) reference, which we defer to revision.

What the DFT data **does** establish is twofold: (i) Cluster I is internally consistent in its disagreement with cheap-DFT (both Orb-v3 OMat and GFN1-xTB anti-correlate with B3LYP/def2-SVP at almost identical r), corroborating that Cluster I is a coherent ranking family; and (ii) Orb-v3 OMol25 is uncorrelated with cheap-DFT ($r \approx +0.09$), so even if OMol25 is “correct” by the standard of ωB97M-V, it is not predicting the cheap-DFT ranking either. We discuss the practical implications of this in Section 4.

3.5 Withdrawn: the pIC50 biology-validation is void (its potency axis is fabricated)

An earlier draft reported that Orb-v3 OMat predicts experimental MMP-1 pIC50 about twice as well as Orb-v3 OMol25 (Spearman $\rho = -0.64$ vs -0.33 across the 15 ligands), and presented this as a biology validation of the cluster paradox. That analysis is withdrawn in full, not merely corrected: the panel’s potency values were subsequently found to be fabricated (§2.1), so the target variable does not exist and no experimental potency for this panel is available. A secondary defect was present even on the file's own numbers — the across-ligand ranking is dominated by a ligand-size confound ($\rho(\text{natoms}, \text{pIC50}) = +0.671$ on the fabricated values; for methods whose absolute energies scale with size the mean energy tracks size, and per-atom normalization flips both GFN1 and AceFF to $\rho \approx +0.23$, the opposite sign) — but the primary reason for withdrawal is that the potency axis is not real.

The withdrawal does not affect the headline cluster-paradox finding, because the per-ligand intra-conformer Pearson is size-invariant by construction: every conformer of a given ligand has the same molecular formula, so all conformer-to-conformer comparisons are at fixed size. The withdrawal is isolated to the across-ligand mean-energy biology validation, which we flag as a methodological pitfall (Discussion §4.5) and which justifies our restriction to per-ligand intra-conformer correlations everywhere else in this manuscript.

Genuine binding-affinity validation of the cluster paradox — i.e., does OMat-ranked top conformer give better Vina/Boltz affinity → better ABFE → better experimental pIC50 reproduction than OMol25-ranked top conformer? — requires full ABFE simulations and is the subject of companion paper_B (Boltz-2x `--use_potentials` protocol) and future paper_A extensions.

3.6 Materials-domain NN sub-cluster (MatterSim × Orb-v3 OMat)

To probe whether the OMol25 paradox is specific to the Orb-v3 architecture or is a property of OMol25 training data, we added MatterSim-5M (Microsoft Research; Yang et al., arXiv: 2405.04967) — an M3GNet-class universal MLIP trained on Materials Project periodic structures — as an 11th axis. MatterSim and Orb-v3 are entirely different architectures (M3GNet vs. Orb-v3's graph-NN gradient model), but they share a training-data domain (materials/inorganic DFT). The cross-replicate Pearson results are:

Pair	Mean ± SD (N=13)	Interpretation
MatterSim ↔ Orb-v3 OMat	**+0.936 ± 0.006**	□ tightest NN–NN pair in our matrix
MatterSim ↔ GFN1-xTB	+0.856 ± 0.011	Materials NN agrees with semi-empirical QM
MatterSim ↔ AIMNet2	+0.750 ± 0.018	Materials NN agrees with

Pair	Mean ± SD (N=13)	Interpretation
		molecular-large NN
MatterSim ↔ AceFF-2	+0.614 ± 0.023	Materials NN agrees with Acellera QM
MatterSim ↔ Orb-v3 OMol25	**+0.410 ± 0.026**	OMol25 paradox confirmed by an independent architecture

Two completely different NN architectures (M3GNet-class MatterSim, gradient-graph Orb-v3) trained on the same domain (materials) give the most-tight NN–NN agreement in our matrix (r = 0.94) — tighter than either's agreement with GFN1-xTB. Yet either of them placed against Orb-v3 OMol25 falls to r = 0.40. This confirms that the OMol25 paradox is a property of the **training-data domain**, not of the Orb-v3 architecture. Companion data from paper_A v4 (Section 3.7) shows the same dataset-determines-cluster pattern for a third architecture (SevenNet-class), making the conclusion robust across at least three NNP architectures.

3.7 Cross-architecture confirmation: SevenNet OMat-trained joins Cluster I; SevenNet OMol25-trained is isolated

In supplemental analysis (paper_A v4, on Boltz seed-49 / v18, 1,244 common ligand poses), we expanded the panel to SevenNet-class universal MLIPs trained on four different datasets: SevenNet-OMat24, SevenNet-MPA, SevenNet-MatPES-PBE, and SevenNet-OMol25-high. The 9-method Spearman ρ matrix (with FeNNix-Bio1S sign-flipped from positive to negative convention) is reported in Table 4.

Table 4. Cross-architecture cluster confirmation (Spearman ρ, v18 n=1,244, paper_A v4).

	xtb	OMat	OMol25	SN-OMol25h	SN-OMat24	SN-MPA	Ma
xtb	+1.000	+0.966	+0.600	+0.603	+0.964	+0.965	+0
OMat	+0.966	+1.000	+0.658	+0.662	+0.999	+0.999	+0
OMol25	+0.600	+0.658	+1.000	+0.996	+0.656	+0.657	+0
SN-OMol25h	+0.603	+0.662	+0.996	+1.000	+0.661	+0.662	+0
SN-OMat24	+0.964	+0.999	+0.656	+0.661	+1.000	+1.000	+
SN-MPA	+0.965	+0.999	+0.657	+0.662	+1.000	+1.000	+

	xtb	OMat	OMol25	SN-OMol25h	SN-OMat24	SN-MPA	Ma
SN-MatPES	+0.967	+0.999	+0.658	+0.662	+1.000	+1.000	+
MatterSim-5M	+0.967	+0.999	+0.659	+0.662	+0.998	+0.998	+0
FeNNix-Bio1S	+0.859	+0.806	+0.424	+0.420	+0.806	+0.807	+0

Reading Table 4: the same SevenNet architecture trained on three different non-OMol25 datasets (OMat24, MPA, MatPES-PBE) gives Spearman $\rho = 1.000$ within Cluster B, and trained on OMol25 gives $\rho = 0.66$ against the rest of Cluster B. This is the most direct possible experiment: hold architecture fixed, vary training data, see whether the cluster placement changes. It does. Training data, not architecture, sets the cluster.

(FeNNix-Bio1S, a reactive QM/FF NN from Sorbonne + Qubit, requires a sign-flip from its native positive-energy convention to a negative one; even after that, it agrees only weakly with the rest of Cluster B at $\rho \approx 0.81$. We retain it for completeness but do not over-interpret its position.)

3.8 Mechanism: OMol25 self-admits weakness on long-range interactions

The mechanistic origin of the OMol25 paradox is in the OMol25 dataset paper itself. Levine et al. (arXiv:2505.08762, OMol25 Dataset, Evaluations, and Models) state in the Evaluations section that OMol25-trained models retain "**large errors**" on three classes of physics: **(i) redox processes**, **(ii) spin transitions**, and **(iii) long-range interactions**. The last is decisive for MMP-1 hydroxamate chemistry, where the ligand-internal energy landscape is dominated by:

1. Partial charge transfer between the hydroxamate `–CONHOH` and Zn^{2+} (long-range electrostatics → coordination-bond ionic character);
2. Intramolecular dipole–dipole and π–π stacking between the warhead-bearing arm and the S1' aryl/heteroaryl groups (long-range);
3. Dispersion between the ligand and the implicit-or-explicit protein environment (long-range).

OMol25's training distribution is 83M small molecules, sampled across 83 elements, with high-level ωB97M-V/def2-TZVPD references. But the sampling distribution of conformations is dominated by near-equilibrium and torsional-scan structures of organic molecules in vacuum; the long-range tail of the Coulombic interaction (and the multipole tail of the dispersion interaction) is precisely the regime where Levine et al. flag training-error growth. MMP-1 hydroxamate–Zn^{2+} coordination is exactly long-range physics.

By contrast, materials-trained NNPs (Orb-v3 OMat, MatterSim-5M, SevenNet OMat24/MPA/MatPES) see periodic crystals with **explicit long-range Coulomb interactions** in their training data (Ewald sums, structure factors, transition-metal coordination environments). This is the dual of “OMol25 trained on near-equilibrium organic geometries”: materials-trained MLIPs are over-exposed to long-range coordination chemistry, not under-exposed.

The cluster topology in Tables 2–4 is then a clean physics story: methods that see long-range coordination physics during training (or by construction, like xTB) form Cluster I; methods that don’t (OMol25-trained variants of any architecture) drift to Cluster Isolated; methods with no electronic structure at all (MMFF94, UFF) anti-correlate from a different (strain-energy) direction and form Cluster II.

3.9 Per-ligand σ_E: Orb-v3 OMol25 has 3× the conformer-energy variance of Orb-v3 OMat

A separate per-ligand metric is the conformer-to-conformer energy standard deviation, σ_E (kcal/mol), within each ligand. This measures how much the NNP “spreads” 100 cofold conformers in its energy landscape. On v18 (paper_A v4):

Method	**mean σ_E (kcal/mol)**
GFN2-xTB	6.82
Orb-v3 OMat	5.91
Orb-v3 OMol25	**16.17**
SevenNet OMol25-high	6.95
SevenNet OMat24	6.26
SevenNet MPA	6.19
SevenNet MatPES	6.15

Orb-v3 OMol25 is the only outlier — three times the mean conformer-energy variance of every other method. This is consistent with the long-range-interaction-error mechanism (small geometric perturbations of the hydroxamate’s coordination geometry produce large false energy variations) and is consistent with the observation that the same Orb-v3 architecture trained on OMat does not show inflated σ_E. The inflation is dataset-specific, and within OMol25 it is architecture-amplified: SevenNet-OMol25-high shows σ_E ≈ 7 kcal/mol, close to the cluster mean, indicating that SevenNet’s E(3)-equivariant transformer regularizes the

OMol25 ranking more aggressively than Orb-v3's gradient-graph network does. Architecture controls σ_E magnitude; dataset controls rank.

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4. Discussion

4.1 Architecture is not destiny; training data is

The standard mental model in the NNP literature has been: "newer/larger/equivariant architectures generalize better." Our 13-replicate result challenges that model on a specific, well-controlled metalloenzyme conformer task. The same Orb-v3 architecture trained on materials data lands in the QM-aligned cluster ($r = 0.89$ with GFN1-xTB); the same Orb-v3 trained on molecular data drifts to $r = 0.37$. The Pearson gap, $\Delta r = 0.51$, is larger than the gap between Orb-v3 OMat and AceFF-2 ($\Delta r \approx 0.22$), larger than the gap between Orb-v3 OMat and ANI-2x ($\Delta r \approx 0.44$), and larger than the gap that distinguishes "QM-like" from "force-field" methods ($\Delta r \approx 0.5$ from Cluster I to Cluster II). Changing the training dataset of a single architecture moves it across the QM/FF cluster divide.

The implication is operational. The choice of NNP for any conformer-ensemble downstream task (pose ranking, ML-DFT cross-check, ABFE conformer reweighting, scoring-function feature) must be reported with both architecture **and** training data, and ideally with cross-NNP validation across at least one method from each domain. Reporting only "we used Orb-v3" or "we used a universal MLIP" obscures which cluster the method belongs to.

4.2 Why materials training helps on a molecular task

This is the counter-intuitive part. The standard intuition is that domain-specialized training improves generalization on the domain; OMol25 was specifically designed for molecular chemistry. Yet on a molecular task (Zn^{2+} inhibitor conformer ranking), the materials-trained variants win. We propose three non-exclusive explanations:

1. **Long-range physics coverage** (Sec. 3.8). Materials training data contains periodic Coulomb sums and explicit transition-metal coordination at scale; OMol25's near-equilibrium organic geometries do not.
2. **Element coverage at the Zn^{2+} site.** OMol25 covers 83 elements, including Zn; but the configurations sampled at Zn in OMol25 are skewed toward small Zn-organic complexes, not

enzyme-active-site geometries with three histidines and a fourth bidentate hydroxamate-bound chelator. Materials training covers Zn in a wide variety of coordination geometries via mineral phases (ZnO, ZnS, zinc-phosphate ferroelectrics) and Zn^{2+} in periodic ionic lattices.
3. **Configurational diversity.** Materials training sets sample force/stress space broadly (perturbations along phonon modes, off-equilibrium snapshots from MD); molecular training sets sample primarily torsional/rotamer modes. For a 100-pose cofold ensemble, off-equilibrium configurations are common, and materials training is better-conditioned for them.

A practical prediction following from this: materials-trained MLIPs should be less accurate than OMol25-trained MLIPs on equilibrium-geometry organic benchmarks (QM9, MD17 equilibria, near-minimum SPICE-2 subsets), and more accurate on metalloenzyme ligand ensembles and other transition-metal-containing chemistry. This is a falsifiable prediction we invite the community to test.

4.3 The honest DFT-reference limitation

Our DFT verdict (Section 3.4) shows Orb-v3 OMat and GFN1-xTB both anti-correlating with B3LYP/def2-SVP at $r \approx -0.84$. This is uncomfortable. We acknowledge two scenarios consistent with our data:

- **Scenario A (cluster-paradox-correct):** Cluster I is correct; cheap B3LYP/def2-SVP without dispersion correction is systematically wrong on Zn^{2+} hydroxamate chemistry (def2-SVP under-corrects the basis-set superposition error on the Zn coordination; lack of D3 misses ligand-internal dispersion). The anti-correlation reflects a known cheap-DFT pathology, not a deficiency of Cluster I. Higher-level reference (ωB97M-V/def2-TZVPD or DLPNO-CCSD(T)) should agree with Cluster I.
- **Scenario B (cluster-paradox-inverted):** OMol25 is correct because OMol25 trained on ωB97M-V/def2-TZVPD (a higher-quality reference than our cheap-DFT verdict). The anti-correlation of OMat + xTB with cheap-DFT is incidental shared bias; OMol25's noisier (per Section 3.9) but correctly-trained landscape is the truth.

We cannot distinguish A from B with our current $N = 25$ cheap-DFT verdict. Higher-level reference calculations are the bottleneck. Until then, we report the cluster bifurcation as a robust descriptive finding ("two clusters of methods exist, with ≈ 68σ statistical significance across 13 replicates"), and we do not claim that any one cluster is the unique ground-truth.

This intellectual honesty is, we believe, the more publishable version of the finding. A reviewer asked to choose between "OMat is right, OMol25 is wrong" (which our $N = 25$ data cannot support) and “Cluster bifurcation exists, mechanism is OMol25’s long-range weakness, definitive ground truth pending” (which our data does support) will, we hope, prefer the latter.

4.4 Practical recommendations for metalloenzyme NNP work

Given the data in hand, we recommend:

1. **Choose materials-trained or broadly-trained MLIPs (Orb-v3 OMat, MatterSim-5M, SevenNet OMat24/MPA/MatPES, AIMNet2) over molecular-only-trained MLIPs (Orb-v3 OMol25, SevenNet OMol25) for Zn^{2+} metalloenzyme conformer scoring.** This recommendation is well-supported by the cross-replicate cluster placement evidence even before the higher-level DFT verdict.
2. **Always cross-validate across at least two NNP families** when the downstream task depends on conformer ranking. A single-method conformer ranking on a metalloenzyme is a methodological degree of freedom.
3. **For drug-discovery workflows specifically (Vina pose ranking, Boltz-2x cofold scoring, ML-FEP), prefer the cheaper semi-empirical xTB GFN1/GFN2 as a sanity check** alongside any NNP scoring step. xTB is in our Cluster I and is cheap.
4. **Report training data, not just architecture.** “Orb-v3” is insufficient; “Orb-v3 OMat” vs. “Orb-v3 OMol25” is the relevant identifier.
5. **For NNP developers: explicitly augment OMol25 training with transition-metal coordination configurations** drawn from materials databases (or fine-tune on a metalloenzyme set like the Protein Data Bank’s metalloprotein-ligand complexes). This is a tractable engineering fix.

4.5 Methodological aside: the size confound and why per-ligand intra-conformer is the right metric

Section 3.5’s withdrawal is a useful object lesson for the broader NNP-validation literature. An earlier analysis attempted to validate the cluster paradox by correlating ligand-mean energies with the panel’s pIC50 values across the 15 ligands, which appeared to confirm the paradox at $\rho = -0.64$ vs -0.33. That analysis is void, because the panel’s potency axis was found to be fabricated (§2.1, §3.5) — there is no experimental potency to validate against. A secondary methodological lesson survives independent of the retracted data: any cross-ligand mean-energy correlation is exposed to a ligand-size confound, since larger ligands have more atoms and therefore more negative total NNP energy, so a correlation that merely tracks size can masquerade as chemistry.

Per-ligand intra-conformer correlations are immune to this confound because every conformer of a fixed ligand has the same molecular formula and therefore the same size contribution to total energy. The size-invariance is a built-in property of the metric, not a post-hoc correction. We recommend it as a default for any cross-NNP benchmarking task that involves heterogeneous ligand size, and we propose that papers reporting NNP-vs-affinity validations on ligand sets with > 30% molecular-weight spread should include explicit size-control checks.

4.6 Limitations and future work

We acknowledge the following limitations:

1. **Single target system (MMP-1).** All 15 ligands chelate the same Zn^{2+} active site. Generalization to other metalloenzymes (carbonic anhydrase, HDAC, MMP-2/9/13, other transition metals) is plausible but unverified. Tier-2 follow-up: replicate the 10-axis NNP panel on CARBONIC ANHYDRASE II (Zn^{2+} + sulfonamide warhead) and HDAC8 (Zn^{2+} + hydroxamate warhead).
2. **Cheap DFT reference.** B3LYP/def2-SVP without dispersion is below modern best practice (Bursch et al., 2022). Higher-level reference at ωB97M-V/def2-TZVPD (~ 30 hours per ligand at N = 25 conformers, feasible) or DLPNO-CCSD(T)/CBS (~ 1 week per ligand) is the recommended definitive validation.
3. **Cofold sampling bias.** Boltz-2x is itself a learned model; the 100-conformer ensemble per ligand may be biased toward Boltz-2x's training distribution. The cross-replicate variation across Boltz seeds 42–54 (SD ≤ 0.038 on most pairs) suggests Boltz seed bias is small, but a non-Boltz conformer source (e.g., RDKit ETKDG followed by xTB-OPT, or AlphaFold3 cofold) would test seed-source robustness.
4. **No protein-ligand binding affinity link.** We have shown that NNP rankings of conformer ensembles differ by training data. We have not shown that this difference propagates to binding-affinity prediction; the withdrawn analysis in Section 3.5 attempted this link against a potency axis later found to be fabricated. Full ABFE simulations using each NNP family's top-ranked conformer (or NNP-weighted ensembles) as starting structures are deferred to companion paper_B / paper_A v6.
5. **Ligand charge ambiguity.** Hydroxamate `-CONHOH` can deprotonate at physiological pH to `-CONHO⁻`; we ran all NNPs with `charge=0` neutral-form unless explicit metadata flagged otherwise. A formal sensitivity analysis to charge state is recommended.

6. **Pipeline-timing measurements include a fair-share regression caveat.** The Section 2.6 chain-stage baselines are intended as a reproducibility envelope for isolated xtb-cascade runs. When the same 24-core / RTX 5090 host runs concurrent ADMET-AI batches at equal scheduling priority, individual chain stages inflate by ≈ 0.34 × (overlap fraction) × (number of concurrent ADMET chains), saturating near +50 % at eight concurrent ADMET chains. Practitioners reproducing the v77–v94 timing baseline should disable concurrent ADMET workloads, or apply the linear correction reported in Section 2.6.

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5. Conclusion

The OMol25 paradox is a single, sharp empirical observation: **the same Orb-v3 architecture trained on the OMol25 molecular dataset disagrees with semi-empirical QM (GFN1-xTB) more strongly than the same architecture trained on the OMat materials dataset does** (Pearson r = 0.374 ± 0.025 vs 0.886 ± 0.009; Δr = 0.512; ≈ 68σ across 13 independent Boltz-2x cofold replicates of 15 structurally diverse MMP-1-active-site ligands). The paradox is robust to architecture (MatterSim-5M, SevenNet, all materials-trained NNPs join Cluster I; SevenNet OMol25-high joins the isolated cluster with Orb-v3 OMol25), to Boltz seed (13 independent replicates, monotone sign), and to ligand structure (all 15 contribute consistently). The mechanism is OMol25's self-reported weakness on long-range interactions (Levine et al., arXiv:2505.08762), which coincides with the dominant physics of Zn^{2+} hydroxamate coordination.

For practitioners, the conclusion is practical: **training-data domain choice for universal MLIPs is more consequential than architecture choice on a Zn^{2+} metalloenzyme conformer task**. Default to materials-trained or broadly-trained MLIPs (Orb-v3 OMat, MatterSim-5M, AIMNet2) when the application involves transition-metal coordination chemistry; cross-validate across at least two NNP families; report training data alongside architecture in methods sections.

For NNP developers, the conclusion is an engineering target: **augment OMol25 training with transition-metal coordination configurations from materials databases or experimental metalloprotein-ligand complexes**. The paradox is, in this sense, a tractable training-data gap, not a fundamental architectural limitation.

Definitive resolution of which cluster matches high-level DFT (ωB97M-V/def2-TZVPD or DLPNO-CCSD(T)/CBS) is deferred to revision; we report the cluster bifurcation as a robust descriptive finding pending higher-level reference.

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Data and code availability

All raw single-point energy CSVs (per axis, per replicate, per ligand, per conformer) and all per-ligand intra-conformer Pearson tables are released at the companion Zenodo deposit (DOI to be assigned at submission). Analysis scripts are in the open `genesis-medicine` repository at `scripts/round27_paperA/`. The Boltz-2x cofold output PDBs (13 replicates × 15 ligands × 100 conformers = 19,500 PDBs per replicate × 13 = 253,500 PDBs total, ≈ 50 GB) are available on request and will be selectively mirrored to Zenodo for the headline 5 ligands.

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(Full DOIs to be filled in during submission; preprint identifiers and arXiv numbers listed here.)

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Supplementary Material

S1. Full per-replicate per-pair Pearson tables (v11–v23)

(See companion `_metadata/paper_a_data_index.md` for paths to per-replicate CSV files. The headline subset of five pairs is given in Table 1 of the main text; the full 10 × 10 × 13-replicate tensor is in `pilot/round27_paperA/cluster_AB_analysis/v5g_13rep_full.npz`.)

S2. NNP version/build details

Method	Software	Version	Checkpoint / weights	URL/DOI
GFN1-xTB	xtb	6.7.0	built-in	grimme-lab/ xtb
GFN2-xTB	xtb	6.7.0	built-in	grimme-lab/ xtb
GFN-FF	xtb	6.7.0	built-in	grimme-lab/ xtb
AceFF-2	aceforce	2024.x	Acellera proprietary	acellera.com
AIMNet2-NSE	aimnet2	2024.x	`aimnet2_b973c_d3_ens`	github.com/ isayevlab/ AIMNet2
ANI-2x	torchani	2.2	`ani2x` ensemble	github.com/ aiqm/ torchani
Orb-v3 OMat	orb-models	0.5.x	`orb_v3_conservative_inf_omat`	github.com/ orbital-materials/ orb-models
Orb-v3 OMol25	orb-models	0.5.x	`orb_v3_conservative_inf_omol`	github.com/ orbital-materials/ orb-models
MatterSim-5M	mattersim	1.1.x	`MatterSim-v1.0.0-5M`	github.com/ microsoft/ mattersim
SevenNet-OMat24	sevenn	0.10.x	`7net-omat-24`	github.com/ MDIL-SNU/ SevenNet
SevenNet-MPA	sevenn	0.10.x	`7net-mpa`	as above
SevenNet-MatPES-PBE	sevenn	0.10.x	`7net-matpes-pbe`	as above
	sevenn	0.10.x	`7net-omol25-high`	as above

Method	Software	Version	Checkpoint / weights	URL/DOI
SevenNet-OMol25-high				
FeNNix-Bio1S	fennix	0.3.x	`Bio1S`	github.com/sorbonne-cnrs/fennix
MMFF94	RDKit	2024.09	`MMFFGetMoleculeForceField`	rdkit.org
UFF	RDKit	2024.09	`UFFGetMoleculeForceField`	rdkit.org

S3. Bug-fix and gotcha notes (for reproducibility)

- **RTX 5090 sm_120 / torch 2.6 kernel mismatch.** Orb-v3 and SevenNet require torch ≥ 2.7.0 cu128 wheels (or `TORCHDYNAMO_DISABLE=1` and CPU fallback) on Blackwell GPUs. See `feedback_rtx5090_sm120_torch_kernel_compat`.
- **Orb-v3 OMol25 silent fail on missing charge/spin.** Atoms must have `atoms.info["charge"]` and `atoms.info["spin"]` populated; otherwise OMol25 silently runs with charge=0/spin=0, which is wrong for ionized hydroxamates and for Zn-containing extracts. See `feedback_orb_v3_omol_charge_spin`.
- **MatterSim per-atom spin requirement.** MatterSim accepts unannotated atoms; no special handling needed for our Zn^{2+} (d^{10} closed-shell).
- **GFN2-xTB SCF divergence on full protein+ligand complex.** ~3000-atom protein+ligand complexes fail xTB GFN1/GFN2 with SIGSEGV (SCF divergence). GFN-FF works. For all NNP single-point evaluations we use the ligand-only extract (heavy atoms + Zn^{2+} if within 6 Å). See `feedback_xtb_gfn2_protein_complex_segfault`.
- **GFN-FF disk explosion.** xTB GFN-FF writes a ~93 MB `gfnff_topo` file per PDB and does not auto-delete; cleanup must be scripted. See `feedback_xtb_gfnff_disk_explosion`.
- **Across-replicate variance is small but not zero.** SD across 13 Boltz seeds is 0.001–0.038 on all pairs; the cluster topology is stable.

S4. Replicate-to-Boltz-seed mapping

Replicate ID	Boltz seed
v11	42
v12	43
v13	44
v14	45
v15	46
v16	47
v17	48

Replicate ID	Boltz seed
v18	49
v19	50
v20	51
v21	52
v22	53
v23	54

S5. DFT reference data (subset)

CSV: `/home/crazat/genesis_medicine/pilot/round27_paperA/dft_b3lyp_d3_verdict.csv`. Headline rows for CHEMBL406 v22:

Conformer	OMat (eV)	OMol25 (eV)	GFN1 (Eh)	B3LYP/ def2-SVP (Ha)	B3LYP-D3(BJ)/ def2-SVP (Ha)
model_98	−213.373	−47856.230	−80.310	−1785.6547	(not run)
model_36	−213.301	−47856.246	−80.303	−1785.6804	−1785.8444
model_82	−213.842	−47858.637	−80.338	−1785.6600	−1785.8256
model_65	−212.954	−47858.066	−80.288	−1785.6814	−1785.8445
model_79	−212.673	−47857.559	−80.274	−1785.6964	−1785.8597

Pearson r against B3LYP/def2-SVP on CHEMBL406 ($n = 5$): OMat $= -0.84$, GFN1-xTB $= -0.84$, OMol25 $= +0.09$ (all marginal at $p \approx 0.07$; cluster claim does not rest on this).

S6. Companion files and analysis pipelines

See `_metadata/paper_a_data_index.md` for a structured index of CSV outputs, analysis scripts, and figure-generation scripts.

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